

FREDERICTON INTL, NB, Canada

WMO: 717000

Lat: 45.869N

Lon: 66.537W

Elev: 68

StdP: 14.66

Time Zone: -4.00 (NAA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.5	-4.5	-20.4	1.8	-7.0	-15.9	2.3	-3.3	25.5	27.7	23.0	26.3	5.3	230	0.598

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	85.9	69.8	82.6	68.1	79.7	66.3	72.3	82.2	70.5	79.2	68.8	76.3	10.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.1	107.2	76.6	67.5	101.2	74.8	65.9	95.8	73.0	36.1	81.9	34.5	79.1	33.0	76.5	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	19.4	17.3	-17.1	92.0	5.2	2.3	-20.8	93.6	-23.8	95.0	-26.8	96.3	-30.5	98.0
			-17.5	75.6	5.0	1.6	-21.1	76.8	-24.1	77.7	-26.9	78.6	-30.5	79.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.3	16.4	19.0	28.7	40.8	52.6	61.8	67.9	66.3	58.5	46.7	35.7	24.1
	DBStd	19.52	11.87	10.96	9.36	6.91	6.58	6.38	4.86	5.14	7.05	7.31	8.84	10.20
	HDD50	4302	1040	869	662	286	44	2	0	0	9	152	436	802
	HDD65	8174	1505	1289	1126	727	389	135	26	46	217	568	879	1267
	CDD50	1871	0	0	1	9	125	355	555	505	265	50	6	0
	CDD65	269	0	0	0	0	4	39	116	86	23	1	0	0
	CDH74	2418	0	0	2	6	107	421	968	711	195	8	0	0
	CDH80	599	0	0	0	1	24	113	237	174	50	0	0	0
	WSAvg	7.6	7.8	8.1	8.9	8.7	8.2	7.3	6.5	6.0	6.6	7.4	7.7	7.9
	PrecAvg	45.3	4.1	3.4	3.8	3.4	3.8	3.3	3.5	3.6	3.7	3.8	4.4	4.6
Precipitation	PrecMax	59.8	8.8	6.5	7.6	6.1	8.3	7.8	6.9	9.4	10.7	7.2	8.1	8.1
	PrecMin	33.3	0.8	1.1	1.1	0.8	0.8	1.0	1.1	0.7	0.8	0.8	1.7	1.9
	PrecStd	6.4	1.7	1.4	1.4	1.3	1.9	1.6	1.4	2.0	1.8	1.7	1.5	1.8
	PrecStd	6.4	1.7	1.4	1.4	1.3	1.9	1.6	1.4	2.0	1.8	1.7	1.5	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.0	48.9	58.7	70.0	83.1	88.4	89.7	89.3	86.3	74.0	62.9	53.1
		MCWB	51.4	45.6	47.4	52.2	64.1	70.2	72.3	71.8	70.6	61.9	59.7	51.6
	2%	DB	44.4	42.2	49.2	62.4	76.4	83.1	85.4	84.7	79.3	67.2	58.5	47.0
		MCWB	43.1	39.3	42.1	49.2	59.8	66.7	70.2	69.7	66.9	59.0	55.1	45.5
	5%	DB	38.1	38.5	45.0	57.4	71.0	78.8	82.5	81.0	74.7	63.0	54.1	42.0
		MCWB	35.9	35.6	39.8	47.8	56.9	64.2	68.6	67.8	64.6	57.3	50.6	39.8
	10%	DB	33.4	35.0	41.4	53.2	66.3	74.7	79.7	77.8	70.9	59.5	49.2	37.7
		MCWB	31.8	32.5	36.5	45.1	54.2	61.9	67.0	66.1	62.8	54.1	46.2	35.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.3	46.3	50.4	58.5	66.6	72.7	75.0	73.9	72.4	65.3	60.8	51.8
		MCDB	51.9	48.5	53.4	64.9	78.4	85.9	85.4	83.6	83.7	69.3	62.6	53.0
	2%	WB	43.2	40.1	44.3	53.2	62.4	69.4	72.6	71.9	69.3	61.7	56.0	46.0
		MCDB	44.2	41.6	47.3	58.7	71.5	79.0	82.0	81.6	75.6	65.1	57.8	46.9
	5%	WB	36.1	35.8	40.3	49.6	59.2	66.9	70.8	70.0	66.9	58.4	51.1	40.2
		MCDB	37.8	37.8	44.0	55.5	67.1	74.7	79.3	77.9	72.1	62.9	53.3	41.5
	10%	WB	32.0	32.7	37.1	45.9	56.3	64.6	69.0	68.1	64.6	54.8	46.2	35.7
		MCDB	33.2	34.7	40.9	52.2	64.3	71.5	76.6	74.9	69.0	58.6	48.8	37.4
Mean Daily Temperature Range	5% DB	MDBR	17.7	18.5	17.2	18.1	20.8	20.8	20.4	20.8	21.0	18.3	14.4	14.8
		MCDBR	23.3	19.1	21.4	26.8	30.8	28.1	25.2	25.6	24.9	24.0	20.2	18.4
		MCWBR	23.0	16.7	15.9	16.7	16.7	14.8	12.3	12.9	14.4	16.7	17.7	17.4
	5% WB	MCDBR	23.5	18.0	19.4	22.6	25.1	23.0	21.5	21.5	20.6	20.8	18.3	18.3
		MCWBR	23.6	16.7	15.9	16.5	15.3	13.5	11.4	11.9	13.1	16.5	17.8	18.2
Clear-Sky Solar Irradiance	taub		0.276	0.286	0.317	0.348	0.374	0.407	0.426	0.400	0.369	0.328	0.306	0.287
	taud		2.453	2.410	2.435	2.420	2.390	2.302	2.269	2.345	2.405	2.525	2.538	2.475
	Ebn at Noon		267	285	291	289	284	273	266	269	268	267	254	249
	Edh at Noon		22	27	31	34	37	40	41	37	32	24	20	19
All-Sky Solar Radiation	RadAvg		475	751	1114	1379	1582	1711	1741	1541	1247	770	453	352
	RadStd		50	67	73	129	181	131	138	98	91	73	52	33

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	+2.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+94	N/A	

Nomenclature: See separate page